AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 33 · STUDENT LESSON

Capstone hypothesis and baseline

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Write the hypothesis. Compute the baseline. Reserve the evidence.

Design a testable capstone comparison without claiming a result before running it.

Keep one habit: Question → data roles → complete baseline → strict criteria → separate pictured proposal → plan audit → independent new target.

MAIN LESSON · SECTION 1

Write a testable plan before inspecting the result

> A hypothesis is a prediction; the experiment supplies the evidence.

Learning goals

I can write a testable capstone **hypothesis** and a credible **baseline**.

I can specify data roles and a selection rule before inspecting results.

I can calculate **baseline** MSE and distinguish a **success criterion** from a result.

Retrieve experiment records

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Similarity and normalisation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- Why must a checkpoint chosen using validation data still face an untouched final test?

> **My starting explanation**
>
> _____
>
> _____
>
> _____
>
> _____

End goal

Explain capstone **hypothesis** and **baseline** with a traceable toy-network case and a bounded conclusion.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

This week you will practise:

Write a testable capstone **hypothesis** and a credible **baseline**

Specify data roles and a selection rule before inspecting results

Calculate **baseline** MSE and distinguish a **success criterion** from a result

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Define better using a model, baseline and metric

> Keep the proposed change small enough to test.

Complete capstone question and prespecified candidate

A capstone needs a question that **evidence** can answer. "Build a better network" leaves better undefined.

Here the question is: does two-update full-batch **fitting** of one scalar ReLU weight reduce validation mean squared error relative to the initial-weight **baseline** on a constructed positive-**input** task?

Predict max(0, a*x), with no bias. The **baseline** has a = 1. The trained candidate starts at a = 1, uses learning rate 0.1 and exactly two updates. These choices are written before results are inspected.

Hypothesis, uncertainty and baseline scope

The **hypothesis** is that the trained candidate has lower validation MSE. The plan does not guarantee that result. This task tests **training** and record keeping, not whether neural networks are preferable to every other model.

A stronger comparison may also include an ordinary fitted line.

Name given and missing evidence

List the given facts and one fact that is not supplied.

MAIN LESSON · SECTION 3

Calculate the baseline and preserve the data roles

> Training fits parameters; validation compares; final cases stay out of choices.

Complete training/validation manifest and reserved role

Role and ID	x	y
Training T1	1	2
Training T2	2	4
Validation V1	3	6
Validation V2	4	8
Final test	Sealed	Sealed

Baseline arithmetic and evaluation rule

Training rows determine updates. Validation rows compare the prespecified candidate with the **baseline**. The final-test **input**s and targets remain sealed until the procedure and reporting plan are frozen.

MSE = sum of squared **prediction** errors divided by the number of evaluated rows. **Baseline prediction**s on V1 and V2 are 3 and 4. Errors are -3 and -4, squared errors 9 and 16, and validation MSE is 25/2 = 12.5.

There are two validation cases.

Every baseline prediction and squared error

Validation ID · Input · Target · Baseline prediction a=1 · Error · Squared error

> **Validation ID:** V1
>
> **Input:** 3
>
> **Target:** 6
>
> **Baseline prediction a=1:** 3
>
> **Error:** -3
>
> **Squared error:** 9

> **Validation ID:** V2
>
> **Input:** 4
>
> **Target:** 8
>
> **Baseline prediction a=1:** 4
>
> **Error:** -4
>
> **Squared error:** 16

> **Validation ID:** MSE denominator
>
> **Input:** Two validation rows
>
> **Target:** (9+16)/2 = 12.5
>
> **Baseline prediction a=1:** Candidate not yet run
>
> **Error:** Final not measured
>
> **Squared error:** No improvement yet established

Follow the three data-role lanes

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Explain the picture

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

State a criterion without calling it a result

> Strict improvement means lower MSE, with equality failing the criterion.

Current task and response

- Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured information leak

- Change: Give only B the eight evaluation labels during **training**.

▼ **Check after predicting**

Prediction: The proposed comparison becomes contaminated rather than a valid held-out test.

Reason: One version has gained privileged information from the reserved evaluation set.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Freeze the comparison and define when evaluation happens

> A written criterion says how a future result will be judged.

Main validation/final criteria and versioned changes

The validation criterion is candidate MSE strictly below **baseline** MSE on V1 and V2. Equality does not pass. The final criterion is lower MSE than the same **baseline** on the sealed final rows.

Report both scores and every raw **prediction** whether or not the criterion passes. Record the rule, **training** rows, update count, learning rate, loss, data versions and code version.

A criterion specifies how to judge a future result; it is not **evidence** that the result already happened.

Keep validation and final criteria separate. Do not silently add epochs or change pre**process**ing after seeing a weak result. A changed plan is a new version and needs an honest account of what **evidence** informed it.

Separate pictured hypothesis and limited supplied facts

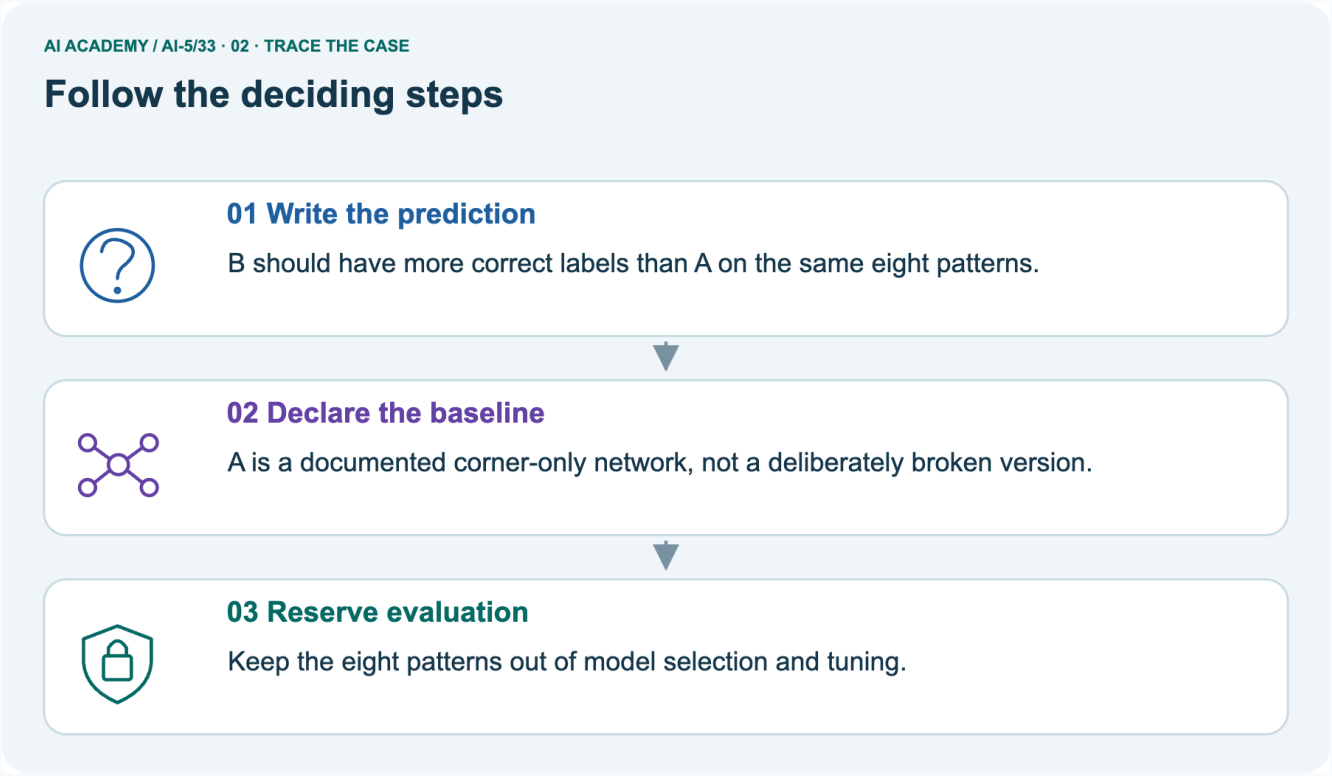
A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

The main capstone is scalar regression with MSE; the picture is a separate proposed image-classification comparison. Pictured A uses corner features; B adds a centre feature. The outcome would be the number correct on the same eight reserved toy patterns.

The pixel grids, model weights, exact labels and predictions are not supplied, so this picture is a planning example, not a runnable scored experiment or evidence that B wins. Both versions must keep reserved labels out of fitting and selection.

Trace change, baseline and measured outcome



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Write the prediction	B should have more correct labels than A on the same eight patterns.
2. Declare the baseline	A is a documented corner-only network, not a deliberately broken version.
3. Reserve evaluation	Keep the eight patterns out of model selection and tuning.

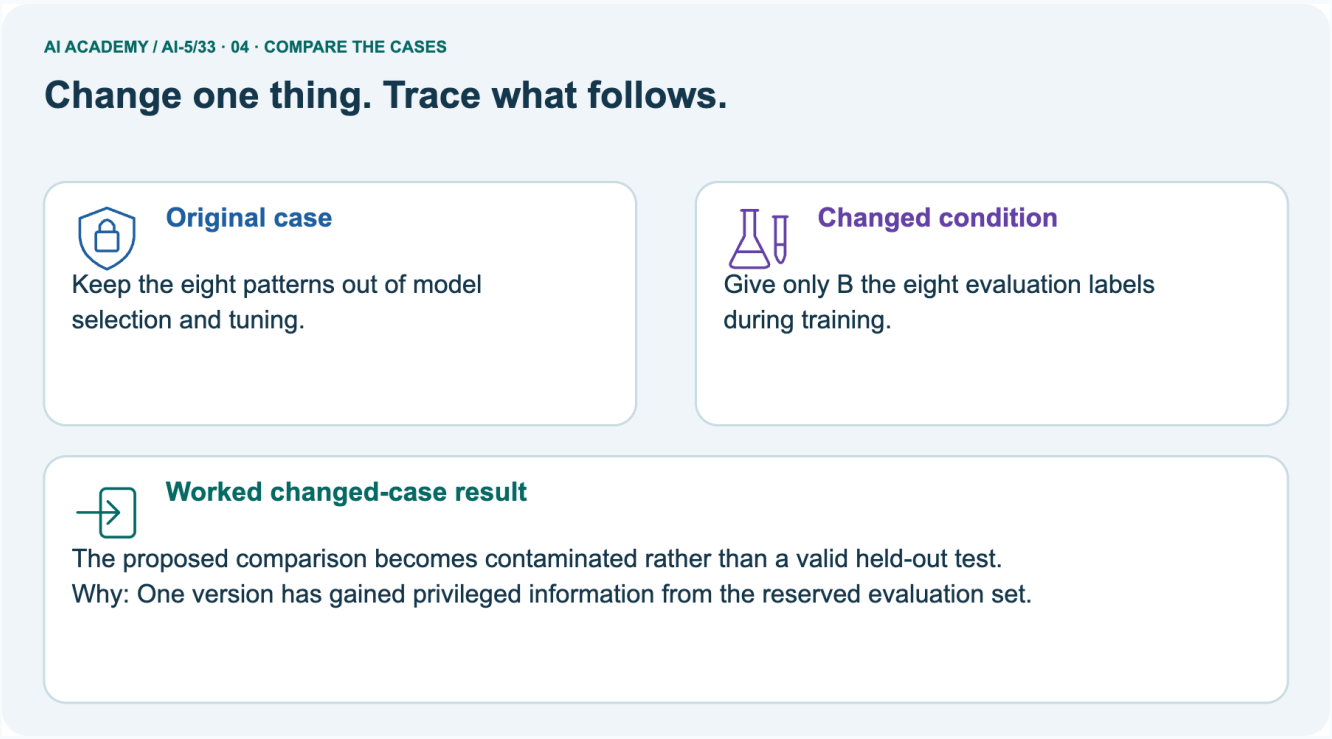
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Predict giving only B the reserved labels

- Change one thing: Give only B the eight evaluation labels during **training**. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Name the intended change and limits

- Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.
- Read the labelled picture: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

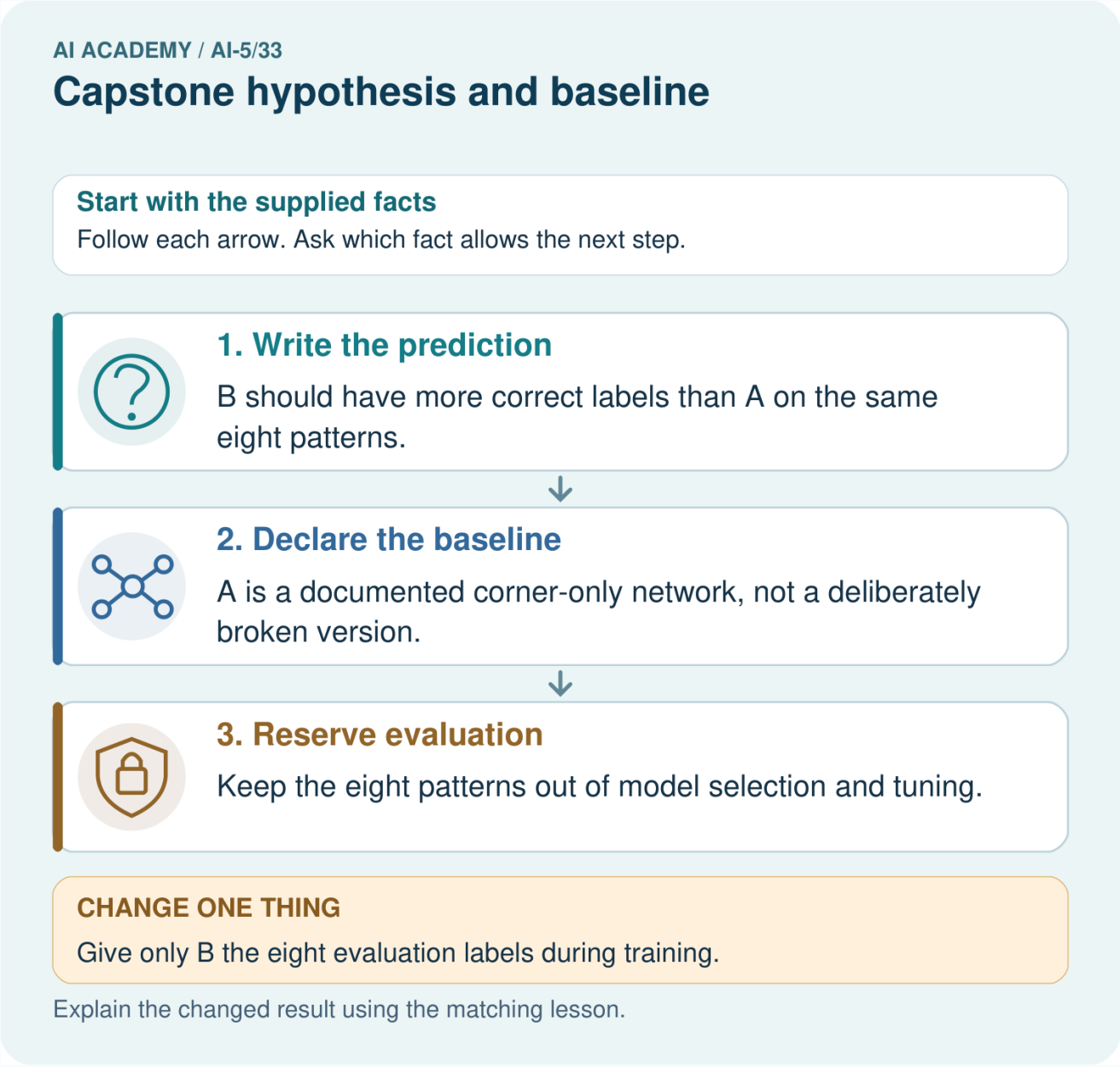
▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone hypothesis and baseline

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Practise identifying what the plan has not measured

> A computed baseline does not supply a candidate prediction.

Complete separate supported baseline and question

- Supported example: a separate **baseline** predicts 2 and 4 for validation targets 5 and 7. Its errors are -3 and -3 and MSE is 9.

  A proposed plan says to train one fixed candidate and report both MSE values on those same two rows. It supplies no candidate **prediction**s, so no improvement can yet be claimed.

Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.

Repair tuning on test outcomes

- A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise a precise hypothesis and comparison contract

> Use the main baseline and the separate supported case with their own numbers.

Current planning task and supported question

- State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.

> **Guided: compute that baseline MSE, name the missing evidence, and state whether the hypothesis has been confirmed.**

> **Explain your prediction to a partner before comparing results.**

Supported W2 record

Use workbook W2 for the connected example “Capstone **hypothesis** and **baseline**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Hypothesis-versus-result check

- Use the worked case for Capstone **hypothesis** and **baseline**. Proposed explanation: “A **hypothesis** is a result”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured comparison response

- Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

> **My explanation using a pictured step and the given facts**

Reserve the independent new target

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Calculate the reference score

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Audit a complete plan without inventing an outcome

> Keep baseline calculations separate from unrun candidate and final results.

Current baseline calculation task

Compute **baseline prediction**s, errors, squared errors and MSE on V1 and V2. State the denominator.

Executable plan and complete baseline rows

```
"""Pre-experiment plan and baseline only; candidate and final results absent."""
import json
plan={'id':'CAP33-v1','question':'Does two-update fitting beat the initial-weight
baseline?',
      'prediction':'max(0,a*x)','bias':False,'baseline_weight':1.0,
      'candidate_initial_weight':1.0,'learning_rate':.1,'updates':2,
      'batch':'full training set','loss':'MSE',
      'validation_rule':'candidate MSE strictly less than baseline MSE',
      'final_rule':'frozen candidate MSE strictly less than same baseline MSE',
      'candidate_result':'not run','final_result':'not measured'}
training=[('T1',1,2),('T2',2,4)]
validation=[('V1',3,6),('V2',4,8)]
ledger=[]
for name,x,y in validation:
    pred=max(0,plan['baseline_weight']*x)
    ledger.append((name,x,y,pred,pred-y,(pred-y)**2))
baseline_mse=sum(r[-1] for r in ledger)/len(ledger)
assert baseline_mse==12.5
```

```
print(json.dumps(plan,indent=2))
print('Training IDs:',[r[0] for r in training])
print('Validation: ID,x,target,prediction,error,squared error',ledger)
print('Baseline validation MSE:',baseline_mse,'denominator:',len(validation))
print('Candidate criterion: MSE <',baseline_mse,'on these same validation rows')
print('Candidate has not been fitted by this script; hypothesis remains untested.')
print('Final inputs/targets intentionally absent from this planning script.')
# Separate supported example, not the main baseline:
assert ((2-5)**2+(4-7)**2)/2==9
```

Expected outputs and honest run status

1. Predict V1/V2 baseline errors and MSE before running or tracing this baseline-only program. It prints the fixed plan, raw baseline rows and denominator 2. It does not train the candidate or open final cases, so candidate success remains untested. Record your actual plan date before your own experiment; this supplied example is not evidence that a real preregistration occurred.

   If you change a candidate or a criterion after inspecting results, preserve the first plan and record the revised version and the evidence that informed it.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: plan for a different target relationship

> Compute the new baseline before writing the candidate criterion.

Complete independent task and reserved final role

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Independent case: Plan a new case independently

Attempt this without hints

A new task predicts max(0, a*x), with **baseline** a = 1. **Training** rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed.

Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates. Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve
</details>

Keep the plan as the first capstone artifact

> Retain the question, baseline, data roles and strict criteria.

Project task and artifact

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A new task predicts max(0, a*x), with **baseline** a = 1.

Training rows are (1,3) and (2,6); validation rows are (3,9) and (4,12); final rows are sealed. Specify a candidate starting at 1 with learning rate 0.1 and two full-batch updates.

Calculate **baseline** validation MSE and write a strict improvement criterion. Has the candidate succeeded yet?

Keep actual **observation**s separate from expected results, preserve failures, compare a **baseline** and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

▼ Earlier wording, alternatives and picture transcripts

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a checkable plan and an unclaimed result

> Someone else should be able to identify every fixed choice.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Write a testable capstone **hypothesis** and a credible **baseline**

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Hypothesis	A testable expectation with a stated comparison and outcome
Baseline	A specified reference procedure evaluated on the same cases
Success criterion	A rule written before evaluation for judging the measured result
Reserved test	Cases kept out of fitting, tuning and selection

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- State the **prediction** rule, **baseline**, candidate learning rate and update count, comparison data and metric. Write a testable **hypothesis**.
- A proposal says “try models until the test score looks good”. Identify the missing fixed choices and explain why that test is not independent.
- A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the new independent target y = 3x planning case for the first individual attempt.

Current explanation and response

- Before running, exchange a plan with a partner. The partner should be able to identify what is changed, what stays fixed, what metric is computed, which rows select the procedure and when the final test may be opened.

  A file name alone does not protect a split; record who can inspect it and how related examples were separated.

- A partner inspects final labels and changes the learning rate. Explain how to document this and obtain independent **evidence**.

> **Exit: someone opens the final targets to choose between learning rates, then calls the winning score independent final evidence. Explain the change of role and the next honest step.**

> **My limited claim and a useful next check**

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
T1 and T2	Fit two updates	Training result only
V1 and V2	Compare with a = 1	Validation comparison
Sealed final rows	Evaluate frozen procedure	Final evidence on those rows

Each arrow preserves a distinct data role. Opening final rows to choose a learning rate changes their role.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

By the end, I can	How I will show it
Write a testable capstone hypothesis and a credible baseline	Explain the deciding step, show a trace or test, and state one limit.
Specify data roles and a selection rule before inspecting results	Explain the deciding step, show a trace or test, and state one limit.
Calculate baseline MSE and distinguish a success criterion from a result	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.
- Which labelled step matches this detail? Keep the eight patterns out of model selection and tuning.
- 1. Reserve evaluation
- 2. Declare the **baseline**
- 3. Write the **prediction**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Reserve evaluation Keep the eight patterns out of model selection and tuning. A capstone **hypothesis** links a specific architectural change to a measured outcome.

A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Changed-case question and optional hint

- Changed-case practice: Give only B the eight evaluation labels during **training**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. The proposed comparison becomes contaminated rather than a valid held-out test.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The proposed comparison becomes contaminated rather than a valid held-out test. One version has gained privileged information from the reserved evaluation set.

Repair and retain the record

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Similarity and normalisation (ai-5/27). Goal: Calculate vector length and normalise a nonzero vector. Short practice: Why do the original two metrics select different winners?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

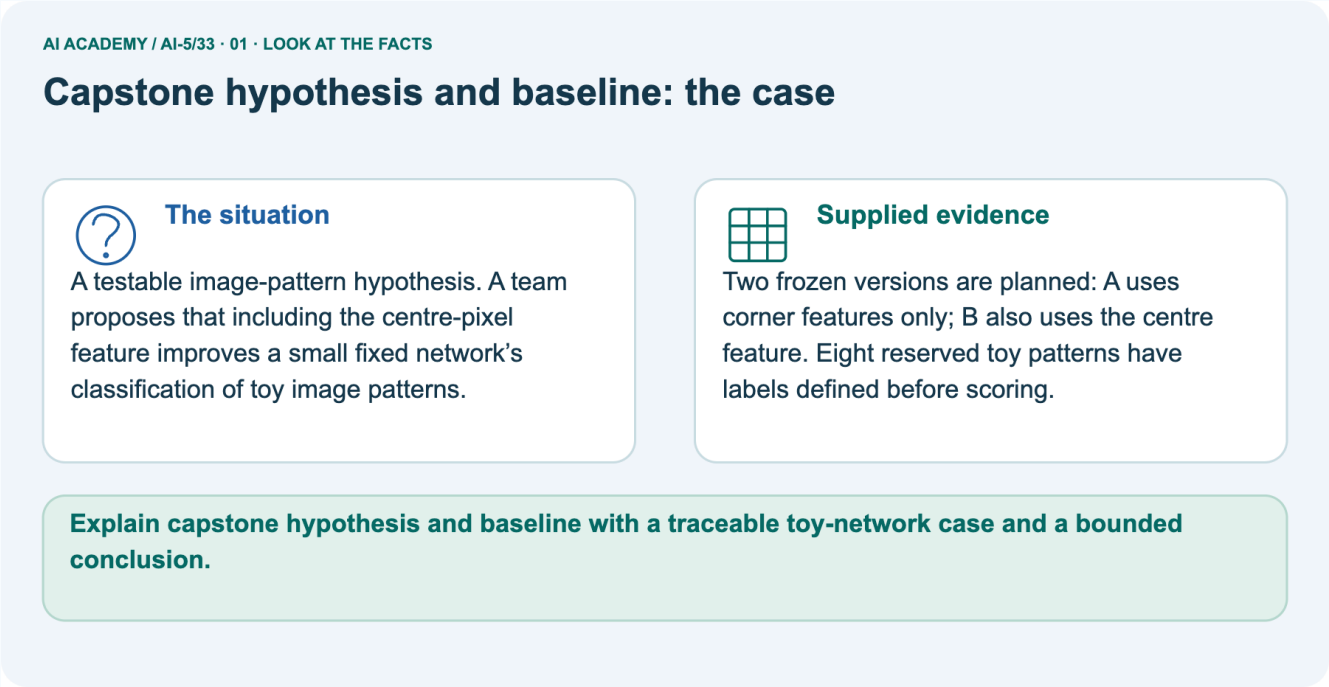
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Capstone hypothesis and baseline: the case



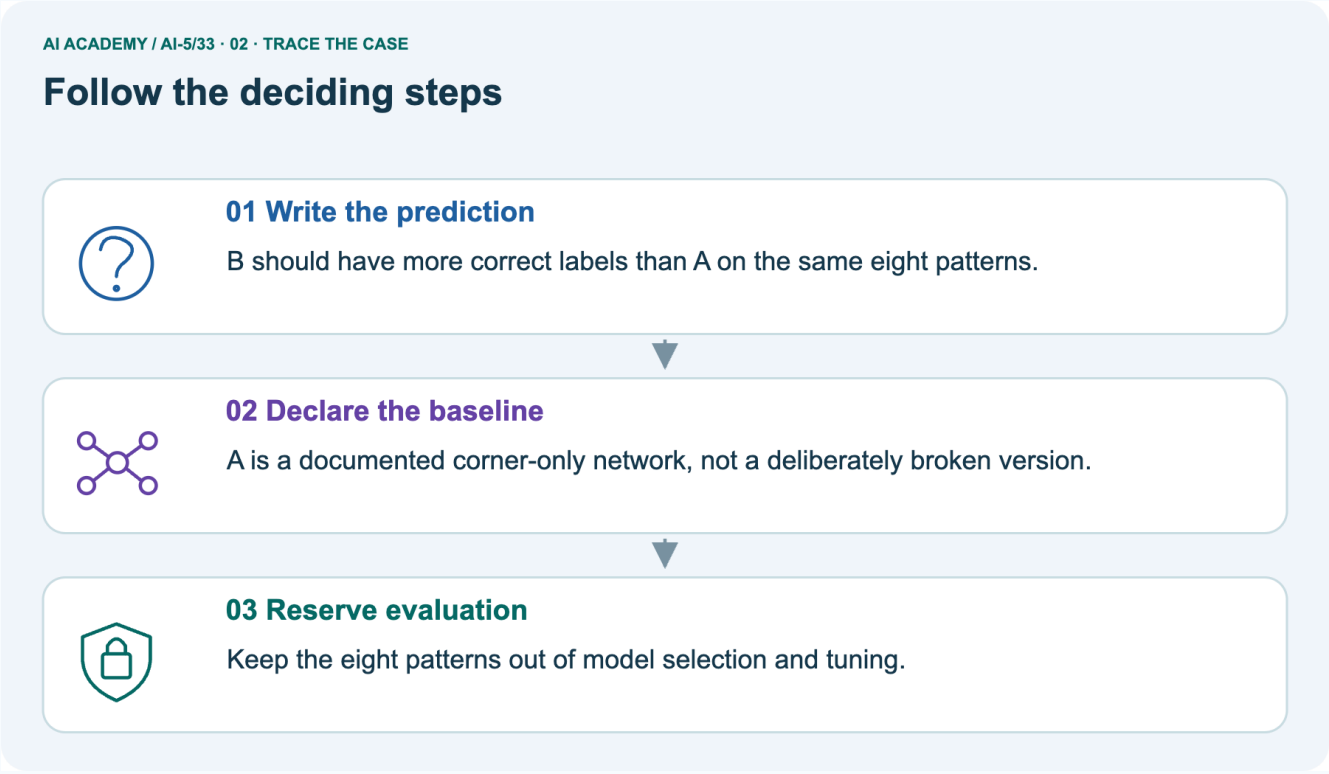
Follow the labelled picture with the matching explanation. Open the vector diagram

A testable image-pattern **hypothesis**. A team proposes that including the centre-pixel feature improves a small fixed network's classification of toy image patterns.

Supplied facts: Two frozen versions are planned: A uses corner features only; B also uses the centre feature. Eight reserved toy patterns have labels defined before scoring.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

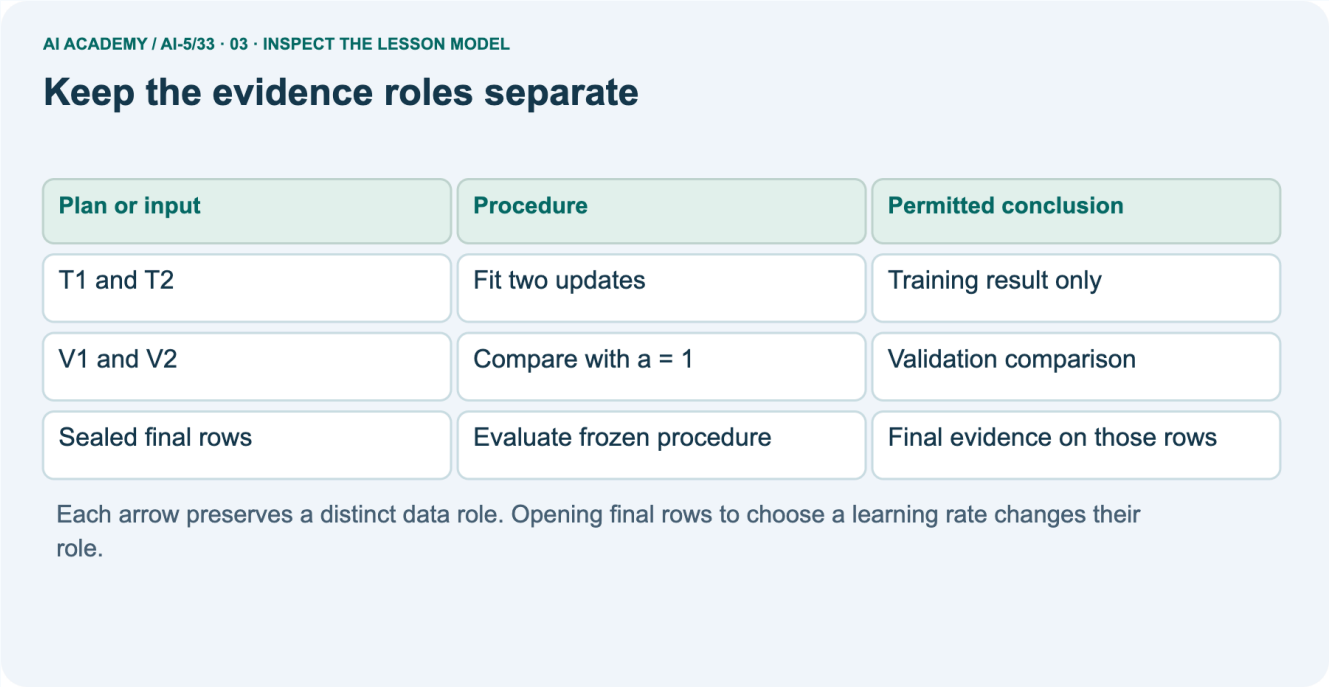


Follow the labelled picture with the matching explanation. Open the vector diagram

Write the **prediction**: B should have more correct labels than A on the same eight patterns. → Declare the **baseline**: A is a documented corner-only network, not a deliberately broken version. → Reserve evaluation: Keep the eight patterns out of model selection and tuning.

Point to each numbered stage and explain which recorded **input** it uses.

03. Keep the evidence roles separate

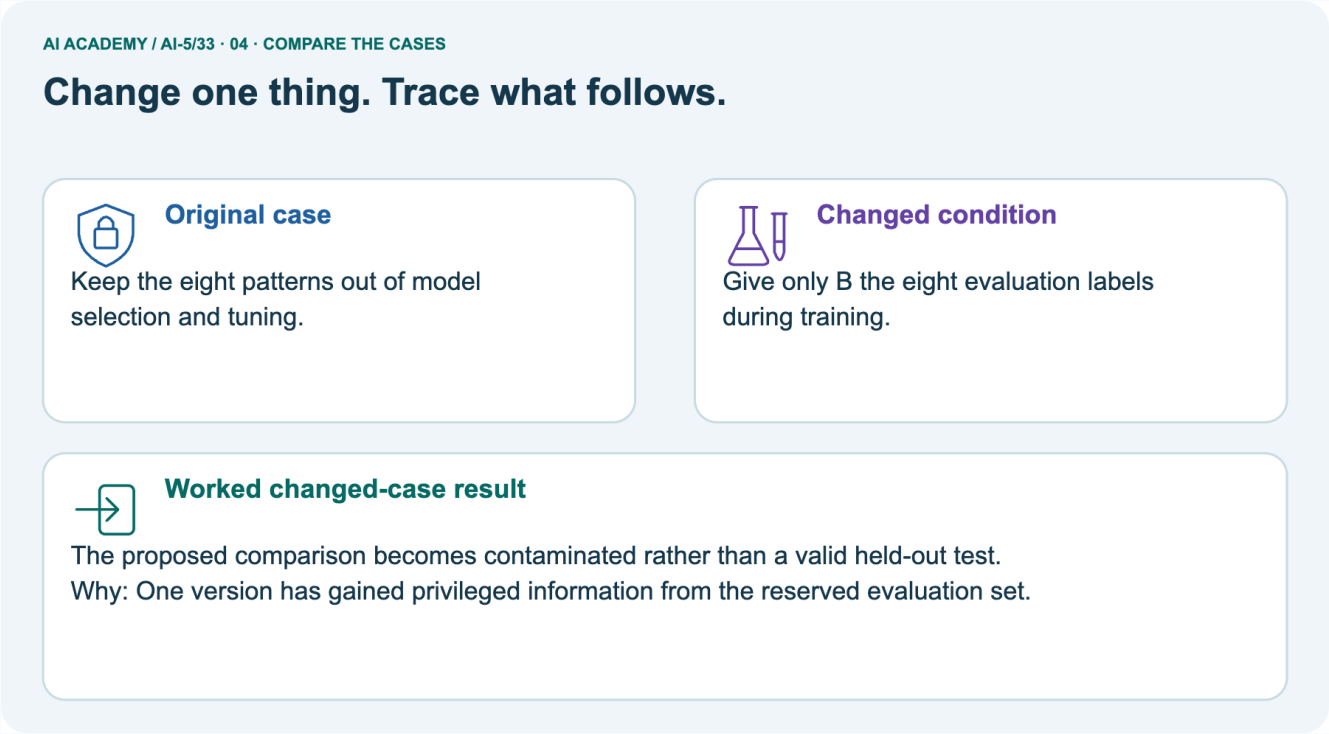


Follow the labelled picture with the matching explanation. Open the vector diagram

Keep the **evidence** roles separate. T1 and T2 | Fit two updates | **Training** result only. V1 and V2 | Compare with a = 1 | Validation comparison. Sealed final rows | Evaluate frozen procedure | Final **evidence** on those rows

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



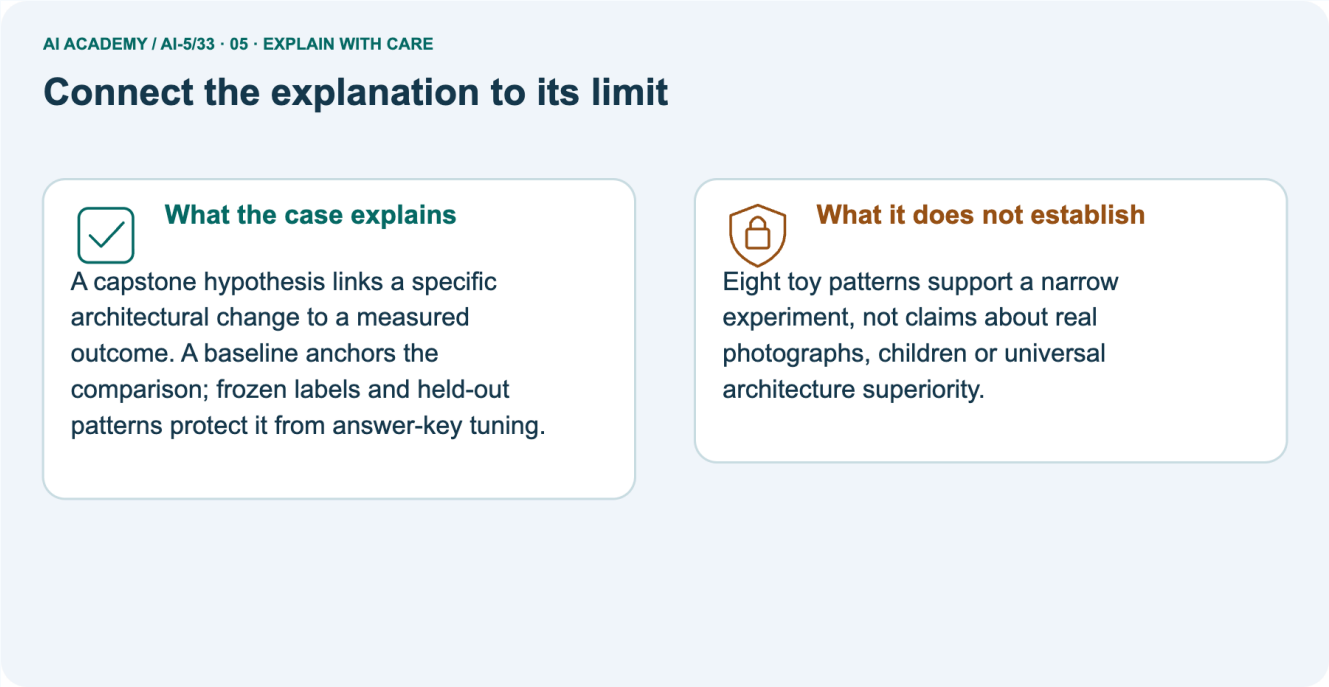
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Keep the eight patterns out of model selection and tuning. Changed condition: Give only B the eight evaluation labels during **training**.

Expected result: The proposed comparison becomes contaminated rather than a valid held-out test. Reason: One version has gained privileged information from the reserved evaluation set.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



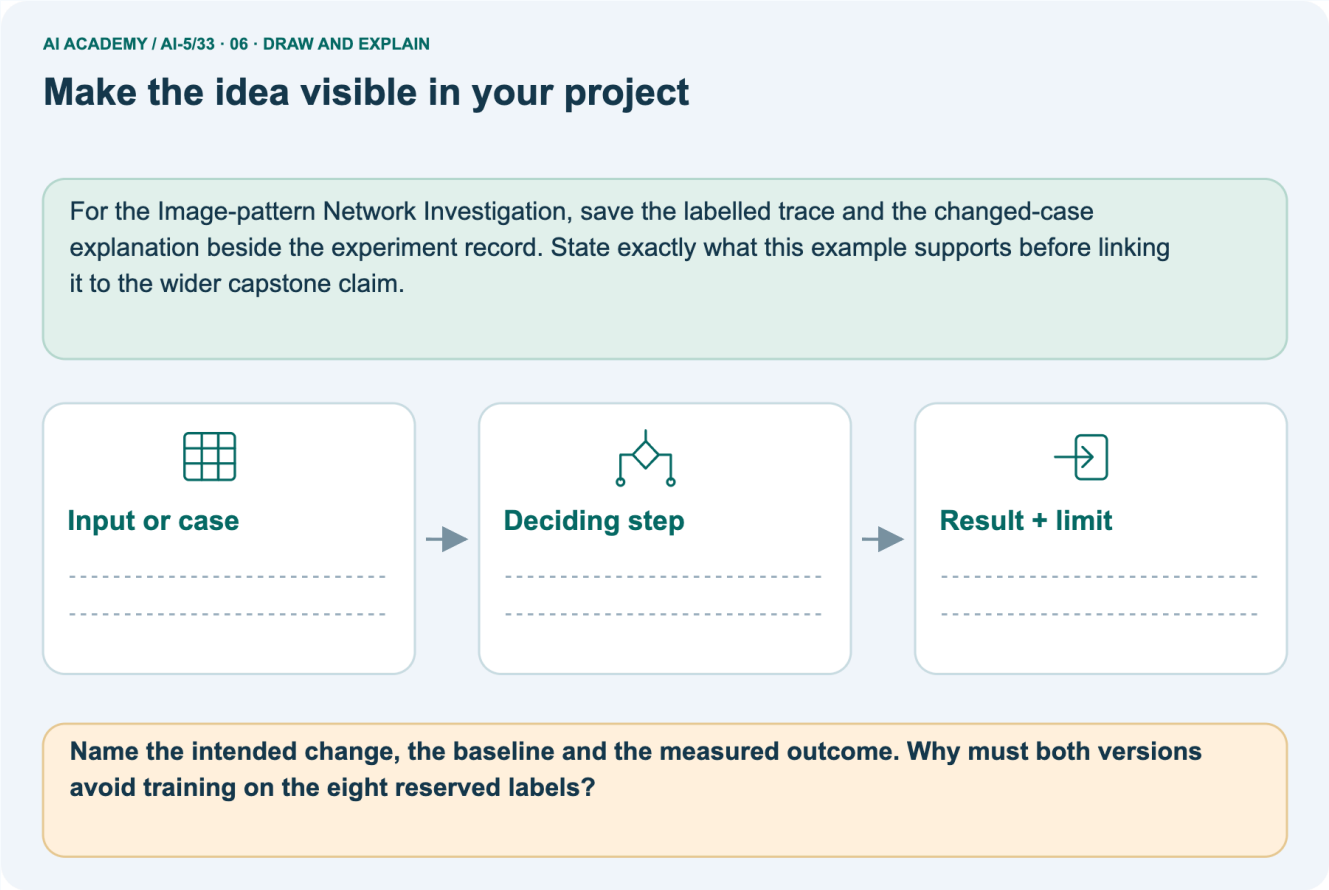
Follow the labelled picture with the matching explanation. Open the vector diagram

A capstone **hypothesis** links a specific architectural change to a measured outcome. A **baseline** anchors the comparison; frozen labels and held-out patterns protect it from answer-key tuning.

Limit: Eight toy patterns support a narrow experiment, not claims about real photographs, children or universal architecture superiority.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Name the intended change, the **baseline** and the measured outcome. Why must both versions avoid **training** on the eight reserved labels?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone hypothesis and baseline: the case

02. Follow the deciding steps

03. Keep the evidence roles separate

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.